

Luminescence Intensity Ratio thermometry and Judd-Ofelt analysis of $\text{TiO}_2\text{:Eu}^{3+}$

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Optical Materials 85 (2018) 261–266

<https://doi.org/10.1016/j.optmat.2018.08.065>

PREPRINT

ABSTRACT

Luminescence thermometry on europium doped TiO_2 films, prepared by the plasma electrolytic oxidation, was investigated by the Luminescence Intensity Ratio method. The performance of the material for temperature sensing was tested from room temperature to 260 °C. The theoretical fit of the emission intensity ratio of Eu^{3+} $^5\text{D}_1 \rightarrow ^7\text{F}_1$ and $^5\text{D}_0 \rightarrow ^7\text{F}_2$ transitions revealed the energy difference between $^5\text{D}_1$ and $^5\text{D}_0$ levels of 1777 cm^{-1} . Temperature sensitivity was calculated on the entire temperature range, with maximum value of 3.43 %K⁻¹ at 293 K. The relationship between Judd-Ofelt and thermometric parameters was demonstrated, and further utilized for the calculation of the radiative transition probability, dipole strength and radiative lifetime of the $^5\text{D}_1 \rightarrow ^7\text{F}_1$ transition.

1. Introduction

Bearing in mind that the temperature is one of the most commonly measured quantities [1], and that temperature sensors comprise 80% of the sensor market [2], the importance of the further thermometry research cannot be overstressed. The absence of the universal readout method and the limits of all the current methods yield the need for finding the appropriate technique for the project specified temperature range and the environment it is read in. For usage in harsh and corrosive environments, or in high electromagnetic fields, the luminescence thermometry has attracted the most attention among the available methods. This semi-invasive method [3] allows single point or 2D thermal imaging even on moving parts or where it is imperative not to disturb the environment of the measurand [4]. The luminescence of lanthanide ions shows narrow emission lines and long emission lifetimes [5], making them an extraordinary material for luminescence thermometry. Luminescence intensity ratio (LIR), is the most used technique for sensing downshifting emissions of rare earth elements (RE). Due to the simplicity and cost effectiveness of the method [6], LIR is often a method of choice for industrial and medicinal uses.

Plasma electrolytic oxidation (PEO) is a novel electrochemical surface treatment process for generating thick oxide coatings on metals, differing from anodization mainly in employing higher potentials than the dielectric breakdown potential [7]. The newly created ceramics greatly improve thermal [8], mechanical and chemical properties

[9,10], which is why the PEO process has found its use in industry and even for medicinal purposes [11]. It is a fast, cost-effective and environmentally friendly process [12]. Additionally, it can be used for doping the metal oxides with RE, thus creating the new materials with good luminescent properties [13].

The luminescence thermometry, photoluminescence, or Judd-Ofelt (JO) analysis of various RE doped oxides has been investigated in great detail, including $\text{TiO}_2\text{:Eu}^{3+}$ [14–16]. However, until this work there has been no luminescence thermometry research or JO analysis on samples prepared by the PEO process. By investigating the thermometric properties, the new functionality is being added to the method already employed in industry. The results obtained here can be used for construction of the novel temperature sensor and protective coatings (e.g. on turbine rotor, or on the surface of airplanes for hypersonic flights), or mark the starting point for luminescence thermometry investigations on other PEO created oxides, possibly doped with other RE ions.

The choice of material for this investigation has fallen on $\text{TiO}_2\text{:Eu}^{3+}$. Titanium dioxide has an excellent chemical, physical, electrical, mechanical and optical properties, high dielectric constant, high melting point, low thermal conductivity, high pH stability, high refractive index, large band-gaps and high transparency in the visible and NIR regions. It is being used as a structural ceramics, in solar cell applications, biocompatible materials or as a catalyst [17–19]. Europium gives very intense luminescence marked with sharp peaks, and is

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unique among the RE in a way that the JO analysis can be obtained from both emission or absorption spectra [20,21]. Regarding the luminescence thermometry utilizing Eu^{3+} -activated materials, emissions from thermally coupled $^5\text{D}_1$ and $^5\text{D}_0$ states are of interest. The energy gap between these states is roughly 1750 cm^{-1} , and is the largest energy gap among all trivalent lanthanide ions used for LIR thermometry. Consequently, intensity ratio thermometry reaches the highest relative sensitivity and the largest high-temperature measurement bound.

These days, of particular interest are materials and structures that have the ability to perform several functions through combinations of structural properties and at least one added functional capability. Therefore, it is rational to expect that with luminescent materials, multifunctionality can be realized by the addition of luminescent thermometric functionality [22]. The added ability of non-contact, remote optical temperature monitoring may be beneficial in contemporary TiO_2 coatings for windows or in solar panels, for example.

The article investigates if the PEO can be used for creating materials suitable for LIR thermometry and is aimed to find the thermometric figures of merit of such materials.

2. Experimental

The Eu^{3+} doped TiO_2 sample was prepared by the PEO method (described in detail in Ref. [23]) from the 99.5% pure (Alfa Aesar) 0.15 mm thick titanium foil of surface dimensions $25\text{ mm} \times 10\text{ mm}$. The pre-processing methods include degreasing in acetone, ethanol and then distilled water, and cleaning ultrasonically. Then the foil was immersed into the electrolyte consisting of water solution 10 g/L sodium phosphate dodecahydrate and 2 g/L Eu_2O_3 powder. The current source set to 150 mA/cm^2 and the oxidizing process was run for 10 min.

X-Ray diffraction (XRD) was used to determine the phase composition of the sample. Measurements were made by Rigaku Ultima IV diffractometer, and data were acquired over the 2θ scattering angle from 20° to 60° with acquisition time of $2^\circ/\text{min}$ and 0.02° step. Rigaku PDLX 2 application software and COD database was used for peak identification.

The complete experimental setup for luminescence thermometry is presented in Fig. 1. The samples were illuminated by the 450 W Xenon lamp. The excitation beam and the resulting radiation is transmitted via an optical fiber bundle. The photoluminescence spectra were recorded by the Horiba Jobin-Yvon Fluorolog-3 Model FL3-221 spectrofluorometer. The samples were placed in a custom-made, temperature controlled, hot-stand. Since the phosphor thermometry must be calibrated against a temperature standard and its accuracy is limited by its reference [24], the reference temperature was measured by the calibrated thermometer.

3. Results and discussion

XRD (Fig. 2) has revealed that the created $\text{TiO}_2\text{:Eu}^{3+}$ coatings consists of anatase (space group $\text{I4}_1/\text{amd}$ [25]) and rutile (space group

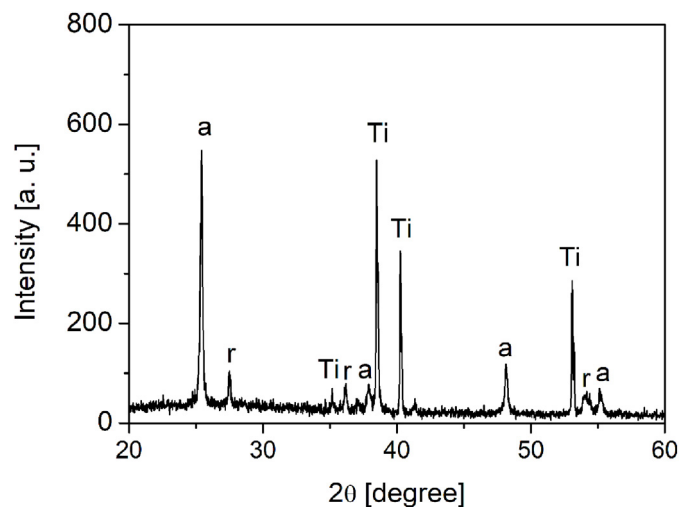


Fig. 2. XRD analysis of TiO_2 obtained by the 10 min PEO process. a – anatase, r – rutile, Ti – substrate.

$\text{P4}_2/\text{mm}$ [26]) phases (the former being the dominant one). The europium species are not visible which is due to the low concentrations. The concentration of incorporated europium species is 0.34% for the 10 min PEO process [15].

The emission spectra were recorded upon the excitation $^7\text{F}_0 \rightarrow ^5\text{L}_6$ with 393 nm beam, over the temperature range from the room temperature to 260°C . On the recorded range 520–640 nm, transitions $^5\text{D}_1 \rightarrow ^7\text{F}_1$ (543 nm), $^5\text{D}_0 \rightarrow ^7\text{F}_0$ (578 nm), $^5\text{D}_0 \rightarrow ^7\text{F}_1$ (590 nm) and $^5\text{D}_0 \rightarrow ^7\text{F}_2$ (613 nm) can be clearly identified [20].

3.1. Luminescence Intensity Ratio

LIR is a ratiometric temperature readout method that utilizes pairs of thermally coupled excited levels from which the emissions with temperature dependent transition intensities occur [6]. Two energy levels are thermally coupled if the energy gap between them is sufficiently small (less than ca. 2000 cm^{-1}) in order to allow the promotion of electrons to the higher state by thermal energy. The states are populated according to the Boltzmann distribution, and since the emission intensities are proportional to the population of each energy level, LIR of two coupled levels (H – higher, L – lower) is given by Refs. [27,28]:

$$\text{LIR}(T) = \frac{I_H}{I_L} = B \exp\left(-\frac{\Delta E}{kT}\right) \quad (1)$$

where $B = A_H \nu_H g_H / A_L \nu_L g_L$, ν_i and g_i are the frequency and the degeneracies of the levels, respectively, ΔE is the energy difference between the two thermally coupled levels, A_i are spontaneous emission rates, and $k = 0.695\text{ cm}^{-1}\text{ K}^{-1}$ is the Boltzmann constant. The value of B depends purely on the material properties, and not on temperature.

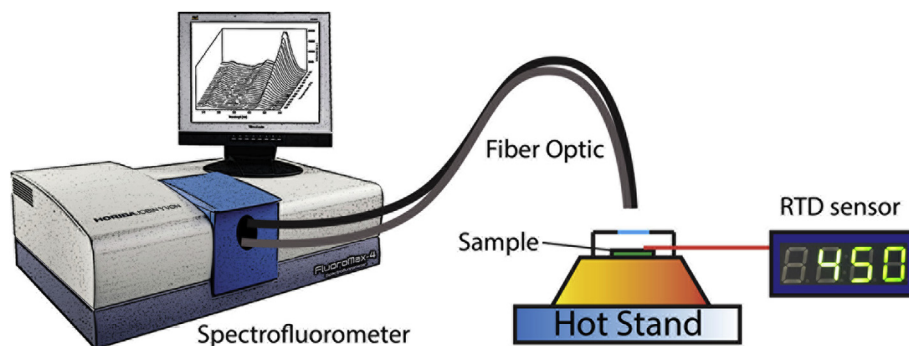


Fig. 1. Scheme of the custom experimental setup used for thermometric measurements.

Quantities that describe the performance of the temperature sensor, absolute and relative sensitivities, can be derived from LIR [29,30]:

$$S(T) = \frac{\Delta E}{kT^2} R(T), \quad S_R(T) = \frac{\Delta E}{kT^2} \cdot 100\% \quad (2)$$

In the case of Eu^{3+} , any combination of emissions ${}^5\text{D}_1 \rightarrow {}^7\text{F}_J$ and ${}^5\text{D}_0 \rightarrow {}^7\text{F}_J$ can be utilized for LIR. Ideal case is to have one emission weakly dependent on temperature to serve as an internal reference [31]. In this case, the hypersensitive transition ${}^5\text{D}_1 \rightarrow {}^7\text{F}_1$ is chosen as an internal reference since the peak at ca. 543 nm has almost constant intensity over the temperature range of measurements. The other, also hypersensitive, transition chosen to measure LIR, ${}^5\text{D}_0 \rightarrow {}^7\text{F}_2$, dominates the spectrum in host matrices with low symmetries and with no inversion center (such is the case of Eu^{3+} doped titanium dioxide) [32,33]. Within the crystal volume they are Ti^{4+} while surface Ti atoms adjust their coordination environment, favoring the formation of Ti^{3+} . Ti^{4+} ions have 6-fold ligand coordination number and are at the center of the D_{2d} distorted octahedron. In anatase titanium dioxide there are three possible sites that can incorporate Eu^{3+} ion: two sites are reduced from the original D_{2d} symmetry to D_2 when Eu^{3+} substitutes Ti^{4+} and to C_{2v} when it substitutes a vacancy, and the third site is with C_1 symmetry at the surface, which occurs when Ti^{3+} is replaced by Eu^{3+} [16,34]. Emission spectra shown in Fig. 3 resemble emission of Eu^{3+} in sites of the lowest symmetry C_1 which are located near nanoparticles' surface or defects and which are, therefore, the most abundant and exist in both anatase and rutile forms of TiO_2 [34]. For this reason, it is impossible to exclude solely from emission spectra emission contribution of Eu^{3+} incorporated in rutile phase; however, this situation doesn't affect thermometry measurements and derived conclusions. One should note from Fig. 3 that in contrast to the ${}^5\text{D}_1 \rightarrow {}^7\text{F}_1$ transition, the ${}^5\text{D}_0 \rightarrow {}^7\text{F}_2$ transition is highly temperature dependent, and its intensity drops rapidly with the temperature increase.

From the fit of the ratio of integrated emission intensities of ${}^5\text{D}_1 \rightarrow {}^7\text{F}_1$ and ${}^5\text{D}_0 \rightarrow {}^7\text{F}_2$ the values $B = 16.2$ and $\Delta E = 1777 \text{ cm}^{-1}$ are obtained (which is in agreement with the value given in Ref. [30]). Theoretical fit together with the relative sensitivity is presented in Fig. 4. The value of relative sensitivity $S_R = 3.43 \text{ \%K}^{-1}$ at 293 K is higher than any sensitivity reported in Ref. [6], see Table 1.

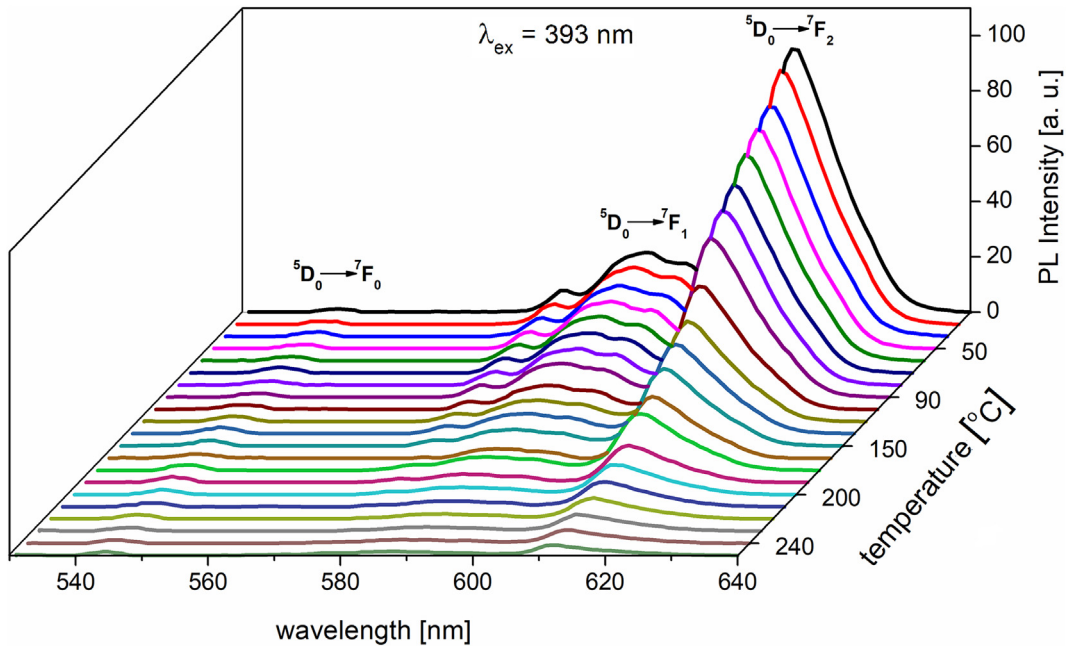


Fig. 3. Emission spectra of the $\text{TiO}_2:\text{Eu}^{3+}$ sample, at temperatures ranging from 20 °C to 260 °C, taken with 20 °C step.

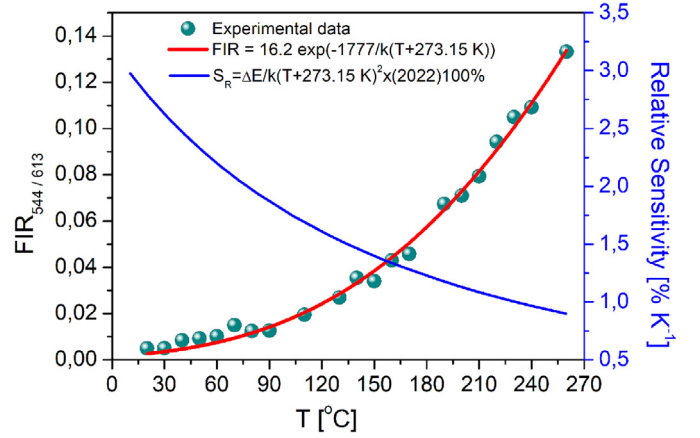


Fig. 4. LIR experimental data, fit and calculated sensitivity.

3.2. Judd-Ofelt analysis

From the absorption or luminescence spectra, JO theory has the ability to predict oscillator strengths, luminescence branching ratios, excited state radiative lifetimes, energy-transfer probabilities and estimates of quantum efficiencies, by using only three parameters. Thus, it has become a centerpiece in RE optical spectroscopy [44]. The europium is unique among the RE elements due to its environment independent magnetic dipole transition ${}^5\text{D}_0 \rightarrow {}^7\text{F}_1$, which can be used as a reference for transitions originating from the ${}^5\text{D}_0$ level [45]. Thus, the JO parameters can be evaluated from the emission spectrum, according to [21]:

$$\Omega_\lambda = \frac{D_{MD} \bar{\nu}_\lambda^3}{e^2 \bar{\nu}_\lambda^3 U^\lambda} \frac{9n_1^3}{n_\lambda(n_\lambda^2 + 2)^2} \frac{I_\lambda}{I_1} \quad (3)$$

where λ labels the total angular momentum of the terminal state, $\bar{\nu}_\lambda$ is the average wavenumber, I_λ is the integrated intensity for transition to ${}^7\text{F}_\infty$, n is the refractive index, D_{MD} is the magnetic dipole strength of the ${}^5\text{D}_0 \rightarrow {}^7\text{F}_1$ transition, and U^λ are reduced matrix elements. The value for $D_{MD} = 9.6 \cdot 10^{-42} \text{ esu}^2 \text{ cm}^2 = 9.6 \cdot 10^{-6} \text{ Debye}^2$ (1 esu = $\text{N}^{-5/2} \text{ cm}$), while all the reduced matrix elements are zero except for: $U^2 = 0.0032$,

Table 1Comparison of thermometric properties of different Eu^{3+} activated materials. Data compiled from Ref. [6].

Material	Range [K]	S_R [%K ⁻¹]	S [K ⁻¹]	ΔE [cm ⁻¹]	Reference
LiNbO ₃	303–723	–	0.0007	–	[35]
Gd ₂ O ₃	300–800	–	0.0007@800 K	1500	[36]
NaEuF ₄	298–523	0.43@293 K	–	1690	[37]
Lu ₂ O ₃	293–823	–	0.0004@823 K	1596	[38]
GdAlO ₃	293–773	2.96@293 K	–	1745	[31]
CaEu ₂ (WO ₄) ₄	300–500	–	0.014@300 K	1068	[39]
Gd ₂ Ti ₂ O ₇	303–423	0.95@423 K	0.015@423 K	1718	[40]
YNbO ₄	303–803	–	0.0035@700 K	1617	[41]
(Y _{0.75} Gd _{0.25}) ₂ O ₃	293–773	–	0.0049@823 K	1633	[42]
Tellurite glass	293–480	2.75@300 K	–	1725	[43]
Calibo glass	293–550	2.75@300 K	–	1725	[43]
GdVO ₄	298–823	–	0.0168@298 K	1530	[28]
TiO ₂ film	293–533	3.43@293 K	–	1777	this work

$U^4 = 0.0023$, $U^6 = 0.0002$ [46]. The JO parameters provide information on the symmetry and bonding of RE within the matrix. Ω_2 is related to the $^5D_0 \rightarrow ^7F_2$ transition, which is related to the covalency and attributed to the increasing polarizability of the ligands around the RE ions [47].

Radiative transition probabilities can be calculated with the following equations [48,49]:

$$A_{\lambda} = \frac{64\pi^4 \bar{\nu}_{\lambda}^3}{3hg} \frac{n_{\lambda}(n_{\lambda}^2 + 2)^2}{9} e^2 \Omega_{\lambda} U^{\lambda}, \quad A_1 = \frac{64\pi^4 \bar{\nu}_1^3}{3hg} n_1^3 D_{MD} \quad (4)$$

where $g = 2J + 1$ is the degeneracy of the initial state.

The radiative lifetime and branching ratios can be calculated according to:

$$\tau_{rad} = \frac{1}{A_R} = \frac{1}{\sum A_{\kappa}}, \quad \beta_{\kappa} = \frac{A_{\kappa}}{\sum A_{\kappa}} \quad (5)$$

and the emission cross-section [50,51]:

$$\sigma_{\kappa}(\lambda_p) = \frac{\lambda_p^4}{8\pi c n_p^2} \frac{\max I_{\kappa}}{\int I_{\kappa} d\lambda} \quad (6)$$

The recorded photoluminescence emission spectrum at room temperature is presented in Fig. 5, with identified sharp emission bands. The $^5D_0 \rightarrow ^7F_6$ intensity was too low to be recorded, thus the Ω_6 parameter could not be determined.

For the refractive index obtained from the Refractive index database

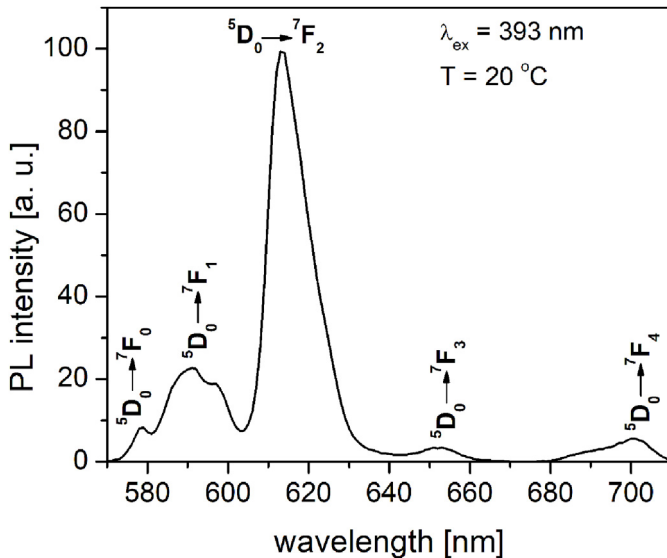


Fig. 5. PL spectrum at room temperature of $\text{TiO}_2:\text{Eu}^{3+}$ created by 10 min PEO process.

Table 2JO intensity parameters and derived quantities obtained from the emission spectrum of $\text{TiO}_2:\text{Eu}^{3+}$, prepared by 10 min PEO process.

κ	1	2	4
$\Omega_{\kappa} 10^{-20}$ [cm ²]	/	4.23	0.42
A_{κ} [s ⁻¹]	200	667	33
β_{κ}	0.22	0.74	0.04
ν_{κ} [cm ⁻¹]	16887	16214	14341
$\sigma_{\kappa} 10^{-22}$ [cm ²]	3.6	16.5	1.3

[52], calculated JO parameters and derived quantities are displayed in Table 2. The total radiative transition rate is $A_R = 900 \text{ s}^{-1}$ and the radiative lifetime equals to $\tau_R = 1.1 \text{ ms}$.

The high Ω_2 parameter is an indication of the low symmetry of the Eu^{3+} site in the host matrix [20]. The high values of the branching ratio and the emission cross section for the $^5D_0 \rightarrow ^7F_2$ transition indicate the possible use of the material for lasing [51].

3.3. LIR and JO

With the data from the LIR analysis and the JO, more information about the hypersensitive $^5D_1 \rightarrow ^7F_1$ transition can be obtained. The transitions $^5D_1 \rightarrow ^7F_1$ and $^5D_0 \rightarrow ^7F_2$ will be abbreviated as H and L , respectively.

The temperature independent constant $B = A_H \nu_H g_H / A_L \nu_L g_L$, by using equation (4) can now be expressed as [53]:

$$B = \frac{n_H (n_H^2 + 2)^2 \nu_H^4}{n_L (n_L^2 + 2)^2 \nu_L^4} \frac{D_H}{e^2 \Omega_L U^L} \quad (7)$$

from where the dipole strength of the $^5D_1 \rightarrow ^7F_1$ transition, depending on the JO parameters and the temperature constant, can be evaluated. By substituting the values obtained for the $\text{TiO}_2:\text{Eu}^{3+}$, the dipole strength of the $^5D_1 \rightarrow ^7F_1$ transition equals to: $D_H = 1.084 \cdot 10^{-42} \text{ esu}^2 \text{ cm}^2$.

With the knowledge of the rate of spontaneous emission of the $^5D_0 \rightarrow ^7F_2$ emission, the rate of spontaneous emission and the radiative lifetime of $^5D_1 \rightarrow ^7F_1$ can be directly calculated:

$$A_H = \frac{B A_L \nu_L g_L}{\nu_H g_H} = 3171 \text{ s}^{-1}, \quad \tau_H = 0.31 \text{ ms} \quad (8)$$

The result indicates very high radiative transition probability, i.e. a fast decay rate, which is in accordance with an assumption that the higher excited states are less stable. The intensity of the $^5D_1 \rightarrow ^7F_1$ transition is given by: $I_H = N_H g_H A_H h \nu_H$, where N_H is the population of the $^5D_1 \rightarrow ^7F_1$ level. The ratio of populations: $N_H / N_L = g_H / g_L \exp(-\Delta E / kT)$ at room temperature is equal to ca. $6 \cdot 10^{-4}$, which explains the low intensity of the $^5D_1 \rightarrow ^7F_1$ transition in the room temperature spectrum of $\text{TiO}_2:\text{Eu}^{3+}$.

4. Conclusion

TiO₂:Eu³⁺ prepared by PEO is a suitable material for temperature sensing. Thus, the new functionality to the already good protective coatings is added, with wide range of applications e.g. in aerospace industry or for temperature readouts on rotors. Our study was first to conduct such measurements on the materials created by PEO.

The experimental data were well fitted to the theoretical model of LIR. LIR method for temperature sensing of the investigated material shows a very high sensitivity for the chosen transitions of Eu³⁺ ions: ⁵D₁→⁷F₁ and ⁵D₀→⁷F₂. The energy difference of the two levels is in agreement with results reported in literature.

JO analysis of the investigated sample has provided the values of the radiative transition probability, branching ratio and emission cross section of transitions originating from ⁵D₀ level. By connecting the LIR and JO, a novel method for calculation of the radiative transition probability, radiative lifetime and dipole strength of the transitions originating from the ⁵D₁ level has been introduced. This extension of the JO theory has been applied on the investigated sample, but can be applied to any Eu³⁺ doped material.

Acknowledgments

This work is supported by the Ministry of Education, Science and Technological Development of the Republic of Serbia under Project No. 171035 and 45020. Authors acknowledge Dr. Sanja Kuzman for spectroscopic measurements.

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